A Dirichlet Pólya Inequality for Three-Dimensional Spherical Shells

Purely analytic main proof manuscript

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Scope and status. This manuscript proves a theorem for the class of genuine three-dimensional spherical shells. It does not claim the general Pólya conjecture, literature novelty, or priority. All shell-specific analytic reductions, boundary assignments, and exact constants are stated in this paper and its accompanying purely analytic supplement. Executable replay sources and interval certificates are retained only as optional regression checks and are not premises of the proof. The imported results are explicitly identified standard Sturm–Liouville and Bessel facts, the published uniform Bessel-phase enclosure, and the annular phase-difference estimate, each quoted with its precise scope.

Abstract

For $0 < r < R$, let

$$A_{r,R} = \{x \in \mathbb{R}^3 : r < |x| < R\}.$$

We prove, with the strict Dirichlet counting convention, that

$$N_D(A_{r,R}, \Lambda) \leq L_3 |A_{r,R}| \Lambda^{3/2}, \qquad L_3 = \frac{1}{6\pi^2}, \qquad \Lambda \geq 0.$$

The proof is purely analytic. It separates variables, converts the radial spectrum into a strict multiplicity-weighted phase count, and combines an exact retained-remainder identity with a ratio-sharp angular block estimate, a tangent-envelope radial minorant, a universal ball quarter-layer, and an elementary scalar convexity argument. The low- and middle-ratio proof starts at the first possible radial frequency; no finite Bessel-zero staircase, event-state theorem, floating-point conclusion, or interval-arithmetic conclusion is a proof premise. An independent appendix gives an elementary proof that no genuine spherical shell tiles $\mathbb{R}^3$ by congruent rigid-motion copies with disjoint interiors, under either exact or almost-everywhere coverage.

Contents

1 Main results and conventions

The eigenvalues of the Dirichlet Laplacian on a bounded domain Ω are listed with multiplicity as

$$0 < \lambda_1^D(\Omega) \le \lambda_2^D(\Omega) \le \cdots .$$

Throughout we use the strict counting function

$$N_D(\Omega, \Lambda) := \#\{j : \lambda_j^D(\Omega) < \Lambda\}. \tag{1.1}$$

This convention matters at every spectral wall. We write $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$.

Theorem 1.1 (Spectral inequality). *For every $0 < r < R$ and every $\Lambda \ge 0$,*

$$\boxed{N_D(A_{r,R}, \Lambda) \le \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} = L_3|A_{r,R}|\Lambda^{3/2}, \qquad L_3 = \frac{1}{6\pi^2}.} \tag{1.2}$$

The standard non-strict formulation follows without changing the constant. Indeed, if

$$N_D^{\le}(\Omega, \Lambda) := \#\{j : \lambda_j^D(\Omega) \le \Lambda\},$$

then discreteness of the Dirichlet spectrum gives

$$N_D^{\le}(\Omega, \Lambda) = \lim_{\varepsilon \downarrow 0} N_D(\Omega, \Lambda + \varepsilon). \tag{1.3}$$

Applying Theorem 1.1 at $\Lambda + \varepsilon$ and passing to the limit proves (1.2) with $N_D^{\le}$ as well. Thus the strict convention used for the wall bookkeeping is equivalent here to the usual non-strict Pólya inequality.

Theorem 1.2 (Non-tiling). *For every $0 < r < R$, no family of congruent rigid-motion copies of $A_{r,R}$ has pairwise-disjoint interiors and covers $\mathbb{R}^3$ exactly or up to a three-dimensional Lebesgue-null set. The same conclusion holds when coverage is formulated using closed shells.*

The two theorems concern literally the same complete shell family. Theorem 1.2 is logically independent of the spectral proof.

Proof map

For $K > 0$, the spectral argument has one global comparison, a no-mode region, and two high-frequency ratio owners. After scaling to $A_{\rho,1}$, separation of variables and the uniform Bessel-phase enclosure give a strict multiplicity-weighted lattice proxy P with

$$N_D(A_{\rho,1}, K^2) \le P.$$

In the action variables of Part I of the supplement, let R be the inverse action, $F = R^2$, and let $M(r)$ count the half-integers strictly below r. The inverse action gives the exact layer-cake picture

$$P = \sum_{\substack{n \ge 1 \\ n - 1/4 < T}} M\big(R(n - 1/4)\big)^2, \qquad W = \int_0^T R(t)^2 \, dt,$$

whose scaled version is Lemma 3.3. Thus the master comparison is

$$W - P = \mathcal{D}_{\mathrm{rad}} - \mathcal{E}_{\mathrm{ang}} : \quad \text{radial quadrature reserve pays angular rounding.} \tag{1.4}$$

The exact definitions and the retained positive remainders in (1.4) are given in the accompanying *Purely Analytic Supplement*.

The parameter space is closed by the following disjoint owners.

frequency range	ratio range	owner
$0 \le K \le \pi/(1-\rho)$	$0 < \rho < 1$	strict no-mode bound
$K > \pi/(1-\rho)$	$0 < \rho < 39/50$	global defect theorem
$K > \pi/(1-\rho)$	$39/50 \le \rho < 1$	optical theorem

Their union proves the unit-shell inequality; dilation proves the result for every $A_{r,R}$.

2 Separated spectrum and phase reduction

Throughout this section

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \qquad 0 < \rho < 1,$$

and the Dirichlet counting function is strict:

$$N_D(A_{\rho,1}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}) < K^2\}.$$

For $x > 0$ we use the strict positive-integer count

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, \lceil x \rceil - 1\}. \tag{2.1}$$

Thus $[m]_< = m - 1$ for $m \in \mathbb{N}$. This convention is retained at every eigenfrequency, phase level, and half-integer angular wall.

2.1 The exact separated spectrum

Let $H_\rho = -\Delta^D_{A_{\rho,1}}$ be the Friedrichs operator associated with

$$\mathfrak{q}[f] = \int_{A_{\rho,1}} |\nabla f|^2 \, dx, \qquad D(\mathfrak{q}) = H_0^1(A_{\rho,1}).$$

Choose an orthonormal spherical-harmonic basis satisfying

$$-\Delta_{\mathbb{S}^2} Y_{\ell m} = \ell(\ell+1) Y_{\ell m}, \qquad \ell \in \mathbb{N}_0, \quad -\ell \le m \le \ell.$$

Proposition 2.1 (Unitary decomposition and completeness)**.** *For*

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2}, \qquad D(L_\ell) = H^2(\rho,1) \cap H_0^1(\rho,1),$$

one has the unitary equivalence

$$H_\rho \simeq \bigoplus_{\ell=0}^{\infty} \bigoplus_{m=-\ell}^{\ell} L_\ell. \tag{2.2}$$

The exact form domain on the right is

$$\left\{ (u_{\ell m}) : \begin{array}{l} u_{\ell m} \in H_0^1(\rho,1), \\ \sum_{\ell,m} \left(\|u_{\ell m}\|_2^2 + \mathfrak{a}_\ell[u_{\ell m}] \right) < \infty \end{array} \right\}, \tag{2.3}$$

where

$$\mathfrak{a}_\ell[u] = \int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2}|u|^2\right) dr.$$

The exact operator domain is

$$\left\{ (u_{\ell m}) : \begin{array}{l} u_{\ell m} \in D(L_\ell), \\ \sum_{\ell,m} \left(\|u_{\ell m}\|_2^2 + \|L_\ell u_{\ell m}\|_2^2\right) < \infty \end{array} \right\}. \tag{2.4}$$

Each L_ℓ has positive, simple eigenvalues tending to infinity and a complete orthonormal eigenbasis. The full operator has compact resolvent.

Proof. Write $x = r\omega$, so $dx = r^2\,dr\,d\omega$, and set

$$R_{\ell m}(r) = \int_{\mathbb{S}^2} f(r,\omega)\overline{Y_{\ell m}(\omega)}\,d\omega, \qquad u_{\ell m}(r) = rR_{\ell m}(r).$$

Fubini, Tonelli, and Parseval on $\mathbb{S}^2$ give

$$\|f\|^2_{L^2(A_{\rho,1})} = \sum_{\ell,m} \int_\rho^1 |R_{\ell m}(r)|^2 r^2\,dr = \sum_{\ell,m}\int_\rho^1 |u_{\ell m}(r)|^2\,dr.$$

Conversely, if $(u_{\ell m})$ is square summable, the finite sums

$$f_L(r,\omega) = \frac{1}{r}\sum_{\ell\le L}\sum_{m=-\ell}^{\ell} u_{\ell m}(r) Y_{\ell m}(\omega)$$

are Cauchy in $L^2(A_{\rho,1})$, and their limit maps back to the given family. Hence the coefficient map is unitary onto the Hilbert direct sum. Because $r, 1/r \in W^{1,\infty}(\rho,1)$, multiplication by either function is bounded on $H^1(\rho,1)$ and preserves the two zero endpoint traces.

For a finite smooth harmonic expansion, polar coordinates and angular orthogonality yield

$$\mathfrak{q}[f] = \sum_{\ell,m}\int_\rho^1 \left(r^2|R'_{\ell m}|^2 + \ell(\ell+1)|R_{\ell m}|^2\right) dr.$$

If $u = rR$, then

$$r^2|R'|^2 = \left|u' - \frac{u}{r}\right|^2 = |u'|^2 - \frac{(|u|^2)'}{r} + \frac{|u|^2}{r^2}.$$

Since

$$\frac{(|u|^2)'}{r} = \left(\frac{|u|^2}{r}\right)' + \frac{|u|^2}{r^2}$$

and $u(\rho) = u(1) = 0$, integration gives

$$\int_\rho^1 r^2|R'|^2\,dr = \int_\rho^1 |u'|^2\,dr.$$

The angular term becomes $\ell(\ell+1)|u|^2/r^2$. Closing finite harmonic expansions in the form norm proves (2.3): one inclusion follows from componentwise form convergence and Fatou's lemma, while the reverse inclusion follows by first truncating the square-summable family in (ℓ,m) and then approximating its finitely many components by $C_c^\infty(\rho,1)$. The uniqueness theorem for operators associated with closed semibounded forms gives (2.2).

For fixed ℓ, the potential $\ell(\ell+1)/r^2$ is continuous and bounded on $[\rho, 1]$. The regular one-dimensional Dirichlet form therefore has associated operator L_ℓ with the displayed $H^2 \cap H_0^1$ domain. The standard domain formula for an orthogonal operator sum gives (2.4).

The embedding $H_0^1(\rho, 1) \hookrightarrow L^2(\rho, 1)$ is compact, so every block has compact resolvent. To pass to the infinite direct sum, note that $r \le 1$ implies

$$L_\ell \ge \ell(\ell+1), \qquad \|(L_\ell + 1)^{-1}\| \le \frac{1}{1 + \ell(\ell+1)}.$$

The resolvent of the full operator is consequently the operator-norm limit of its finite-ℓ truncations, each of which is compact. Hence the full resolvent is compact.

Regular Dirichlet Sturm–Liouville theory gives a complete sequence of simple eigenvalues in every block. They are strictly positive because

$$\mathfrak{a}_\ell[u] \ge \int_\rho^1 |u'|^2 \, dr \ge \frac{\pi^2}{(1-\rho)^2} \|u\|_2^2.$$

Simplicity also follows directly: two solutions of the same second-order equation that vanish at $r = \rho$ have proportional initial derivatives and hence are proportional by uniqueness. If $u_{\ell,n}$ is a complete radial eigenbasis, then

$$\frac{u_{\ell,n}(r)}{r} Y_{\ell m}(\omega)$$

over all (ℓ, m, n) is exactly the inverse image of a complete orthonormal coordinate basis of the Hilbert direct sum. This proves global completeness. $\square$

Proposition 2.2 (Bessel determinant and strict phase enumeration)**.** *Put $\nu = \ell + \frac{1}{2}$. The positive eigenfrequencies in the ℓ-th radial block are precisely the positive zeros of*

$$D_{\nu,\rho}(k) := J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k). \tag{2.5}$$

Every such zero is simple. With the continuous phase branch

$$J_\nu(x) + iY_\nu(x) = M_\nu(x) e^{i\theta_\nu(x)}, \qquad M_\nu(x) > 0, \qquad \theta_\nu(0+) = -\frac{\pi}{2}, \tag{2.6}$$

and

$$\Psi_{\nu,\rho}(K) = \theta_\nu(K) - \theta_\nu(\rho K), \qquad \gamma_{\rho,K}(\nu) = \frac{\Psi_{\nu,\rho}(K)}{\pi},$$

the exact strict count is

$$N_D(A_{\rho,1}, K^2) = \sum_{\ell=0}^{\infty} (2\ell+1) [\gamma_{\rho,K}(\ell + \tfrac{1}{2})]_<. \tag{2.7}$$

The sum is finite. If K is a radial eigenfrequency in one or several angular channels, the root at K is excluded in each such channel and the excluded angular multiplicities add.

Proof. For $k > 0$, the physical radial equation has the general solution

$$R(r) = r^{-1/2} \big(c_1 J_\nu(kr) + c_2 Y_\nu(kr) \big).$$

The two Dirichlet conditions admit a nonzero coefficient vector exactly when

$$\det \begin{pmatrix} J_\nu(\rho k) & Y_\nu(\rho k) \\ J_\nu(k) & Y_\nu(k) \end{pmatrix} = D_{\nu,\rho}(k) = 0.$$

The Wronskian

$$J_\nu(x)Y_\nu'(x) - J_\nu'(x)Y_\nu(x) = \frac{2}{\pi x} \tag{2.8}$$

shows that the two entries in either row cannot vanish simultaneously. Thus the determinant criterion remains exact even if the J_ν values, or the Y_ν values, happen to vanish at both endpoints.

For root simplicity define the left-normalized solution

$$s_\ell(r,k) = \frac{\pi\sqrt{\rho}}{2}\sqrt{r}\left[J_\nu(\rho k)Y_\nu(kr) - Y_\nu(\rho k)J_\nu(kr)\right]. \tag{2.9}$$

Equation (2.8) gives

$$s_\ell(\rho,k) = 0, \qquad \partial_r s_\ell(\rho,k) = 1, \qquad s_\ell(1,k) = \frac{\pi\sqrt{\rho}}{2}D_{\nu,\rho}(k).$$

Let $v = \partial_k s_\ell$. Differentiating $(L_\ell - k^2)s_\ell = 0$ gives

$$(L_\ell - k^2)v = 2ks_\ell, \qquad v(\rho,k) = \partial_r v(\rho,k) = 0.$$

At a positive determinant root, Lagrange's identity yields

$$\partial_k s_\ell(1,k)\,\partial_r s_\ell(1,k) = 2k\int_\rho^1 s_\ell(r,k)^2\,dr > 0. \tag{2.10}$$

Thus $\partial_k s_\ell(1,k) \neq 0$, and the zero of the determinant is simple.

At zero energy the left-normalized solution is

$$s_\ell(r,0) = \frac{\rho^{-\ell}r^{\ell+1} - \rho^{\ell+1}r^{-\ell}}{2\ell+1},$$

so

$$s_\ell(1,0) = \frac{\rho^{-\ell} - \rho^{\ell+1}}{2\ell+1} > 0.$$

Equivalently, the standard small-argument expansions give

$$\lim_{k\downarrow 0} D_{\nu,\rho}(k) = \frac{\rho^{-\nu} - \rho^{\nu}}{\pi\nu} > 0. \tag{2.11}$$

Hence the limiting phase level zero is not an eigenfrequency.

From (2.6),

$$J_\nu = M_\nu\cos\theta_\nu, \qquad Y_\nu = M_\nu\sin\theta_\nu,$$

and direct substitution gives the sign-sensitive factorization

$$D_{\nu,\rho}(k) = M_\nu(\rho k)M_\nu(k)\sin\Psi_{\nu,\rho}(k). \tag{2.12}$$

The Wronskian also gives

$$\theta_\nu'(x) = \frac{2}{\pi x M_\nu(x)^2} > 0.$$

Nicholson's integral representation [3, Eq. (10.9.30)]

$$M_\nu(x)^2 = \frac{8}{\pi^2}\int_0^\infty \cosh(2\nu t)K_0(2x\sinh t)\,dt$$

and the strict decrease of K_0 imply that $M_\nu(x)^2$ is strictly decreasing in x. Therefore

$$\Psi_{\nu,\rho}'(k) = \frac{2}{\pi k}\left(M_\nu(k)^{-2} - M_\nu(\rho k)^{-2}\right) > 0. \tag{2.13}$$

The branch condition gives $\Psi_{\nu,\rho}(0+) = 0$, while the standard large-argument Bessel expansion gives

$$\Psi_{\nu,\rho}(k) = (1-\rho)k + O(k^{-1}) \longrightarrow \infty$$

for fixed ν and ρ. Hence the positive determinant roots are uniquely indexed by

$$\Psi_{\ell+1/2,\rho}(k_{\ell,n}) = n\pi, \qquad n = 1, 2, \dots .$$

Strict monotonicity gives

$$k_{\ell,n} < K \iff n\pi < \Psi_{\ell+1/2,\rho}(K).$$

The direct-sum decomposition supplies angular multiplicity $2\ell + 1$. If the same numerical eigenvalue occurs at distinct values of ℓ, the corresponding spherical-harmonic sectors are orthogonal and their multiplicities add. Finally, $\lambda_{\ell,n} \geq \ell(\ell+1)$, so only finitely many angular channels can contribute below K^2. These observations prove (2.7), including every strict spectral wall. □

The only standard inputs in the preceding proof are completeness of the spherical harmonics, regular one-dimensional Dirichlet Sturm–Liouville theory, the Bessel equation, the Wronskian, Nicholson's representation, and the usual small- and large-argument Bessel formulas; see [3].

2.2 Uniform phase enclosures and the strict lattice proxy

For $a > 0$, define

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \leq z \leq a, \\ 0, & z > a. \end{cases} \tag{2.14}$$

For $0 \leq z < a$, put

$$H_a(z) = \frac{3a^2 + 2z^2}{24\pi(a^2 - z^2)^{3/2}},$$

and define

$$\mathcal{F}_a(z) = \begin{cases} \max\{G_a(z) - H_a(z), -1/4\}, & 0 \leq z < a, \\ -1/4, & z \geq a. \end{cases} \tag{2.15}$$

We use the following published Bessel-phase result, stated here with every hypothesis needed in the sequel.

Lemma 2.3 (External uniform Bessel-phase enclosure). *With the branch* (2.6), *for every* $a > 0$ *and real* $z \geq 0$,

$$\mathcal{F}_a(z) - \frac{1}{4} < \frac{\theta_z(a)}{\pi} < G_a(z) - \frac{1}{4}. \tag{2.16}$$

If $0 < \mu < K$, $0 \leq z \leq K$, *and*

$$\Gamma_{K,\mu}(z) := \frac{\theta_z(K) - \theta_z(\mu)}{\pi},$$

then

$$0 < \Gamma_{K,\mu}(z) < G_K(z) - \mathcal{F}_\mu(z) \leq G_K(z) + \frac{1}{4}. \tag{2.17}$$

If $0 \leq z \leq \mu$, *including* $z = \mu$, *then also*

$$G_K(z) - G_\mu(z) < \Gamma_{K,\mu}(z) < G_K(z) - G_\mu(z) + \frac{1}{4}. \tag{2.18}$$

No value of the divergent expression $H_\mu(\mu)$ *is used in the last statement.*

Lemma 2.3 is Theorem 5.1 and Lemma 5.2 of [2], based on the real-order phase enclosures of [1]. The upper bound in (2.17) follows immediately by subtracting the lower bound for $\theta_z(\mu)$ from the upper bound for $\theta_z(K)$ in (2.16); the correlated two-sided estimate (2.18) is the sharper part of the annulus result. Both theorems are stated for real order, so the half-integers $z = \ell + \frac{1}{2}$ require no interpolation.

Fix now $K > 0$, set $\mu = \rho K$, and define

$$A_{\rho,K}(z) = G_K(z) - G_\mu(z). \tag{2.19}$$

For $0 \le z \le K$, let

$$s_\mu(z) = \begin{cases} \min\{H_\mu(z), 1/4\}, & 0 \le z < \mu, \\ 1/4, & \mu \le z \le K, \end{cases} \qquad U_{\rho,K}(z) = A_{\rho,K}(z) + s_\mu(z). \tag{2.20}$$

The second line, rather than H_μ, is always used at $z = \mu$.

Proposition 2.4 (Strict phase-to-lattice reduction)**.** *For $0 < \rho < 1$ and $K > 0$,*

$$\gamma_{\rho,K}(z) < U_{\rho,K}(z) \le A_{\rho,K}(z) + \frac{1}{4}, \qquad 0 \le z \le K. \tag{2.21}$$

Because every G_a is zero-extended, $A_{\rho,K}$ is the piecewise action $G_K - G_{\rho K}$ below the inner interface and G_K at and above it. Consequently,

$$N_D(A_{\rho,1}, K^2) \le \sum_{\nu_\ell < K} 2\nu_\ell [U_{\rho,K}(\nu_\ell)]_< \tag{2.22}$$

$$\le S_{\rho,K} := \sum_{\nu_\ell < K} 2\nu_\ell \left\lfloor A_{\rho,K}(\nu_\ell) + \frac{1}{4} \right\rfloor, \qquad \nu_\ell = \ell + \frac{1}{2}. \tag{2.23}$$

The cutoff is strict: an order $\nu_\ell = K$ is inactive. At a true phase wall the radial count is one less than the ordinary floor; at a proxy wall the ordinary floor in (2.23) is retained. Moreover,

$$\int_0^\infty 2z A_{\rho,K}(z)\, dz = \frac{2}{9\pi}(1 - \rho^3) K^3. \tag{2.24}$$

Proof. For $0 \le z < \mu$, the first upper bound in (2.17) reads

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + G_\mu(z) - \mathcal{F}_\mu(z).$$

From (2.15),

$$G_\mu(z) - \mathcal{F}_\mu(z) = \min\{H_\mu(z), G_\mu(z) + 1/4\}.$$

The upper half of (2.18) also gives $\gamma_{\rho,K}(z) < A_{\rho,K}(z) + 1/4$. Taking the better of these two strict upper bounds, and using $G_\mu(z) + 1/4 \ge 1/4$, gives

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + \min\{H_\mu(z), 1/4\} = U_{\rho,K}(z).$$

For $\mu \le z \le K$, one has $G_\mu(z) = 0$, and (2.17) gives

$$\gamma_{\rho,K}(z) < G_K(z) + 1/4 = U_{\rho,K}(z).$$

Orders $z \ge K$ do not contribute. Indeed, (2.16) gives

$$\frac{\theta_z(K)}{\pi} < -\frac{1}{4}, \qquad \frac{\theta_z(\mu)}{\pi} > -\frac{1}{2},$$

because $G_K(z) = 0$ and $\mathcal{F}_\mu(z) = -1/4$. Together with phase monotonicity this yields

$$0 < \gamma_{\rho,K}(z) < \frac{1}{4} < 1,$$

so there is no positive phase level below K. This includes $z = K$.

For positive $x < y$, every positive integer strictly below x is also strictly below y, so $[x]_< \le [y]_< \le \lfloor y \rfloor$. Applying this observation to the exact phase count (2.7), and then using $s_\mu \le 1/4$, proves (2.22)– (2.23) without changing either wall convention.

Finally, scaling $z = ax$ gives

$$\int_0^a 2zG_a(z)\,dz = \frac{2a^3}{\pi}\left(\int_0^1 x\sqrt{1-x^2}\,dx - \int_0^1 x^2 \arccos x\,dx\right).$$

The two elementary integrals are $1/3$ and $2/9$, respectively; hence the last expression is $2a^3/(9\pi)$. Subtracting the result with $a = \mu = \rho K$ from that with $a = K$ proves (2.24). $\square$

3 Product and complementary-action tools

We now prove the Pólya inequality uniformly as the shell becomes thin. In this section the counting function is strict:

$$N_D(\Omega, K^2) = \#\{j : \lambda_j(\Omega) < K^2\}.$$

We retain the global convention $[u]_< := \#\{n \in \mathbb{N} : n < u\} = \max\{0, \lceil u \rceil - 1\}$. Thus a spectral level lying exactly on the wall K^2 is not counted. Put

$$\varepsilon = 1 - \rho, \qquad a = \varepsilon K, \qquad m_\varepsilon = 1 - \varepsilon + \frac{\varepsilon^2}{3}. \tag{3.1}$$

Since $1 - \rho^3 = 3\varepsilon m_\varepsilon$, the desired right-hand side has the two equivalent forms

$$\frac{2}{9\pi}(1-\rho^3)K^3 = \frac{1}{\varepsilon^2}\frac{2a^3}{3\pi}m_\varepsilon. \tag{3.2}$$

The proof uses the optimized phase estimate already established in Proposition 2.4. We record precisely the consequence that is used here. For $\nu_\ell = \ell + \frac{1}{2}$, let $\gamma_{\rho,K}(\nu_\ell)$ be the normalized radial Bessel phase. If $0 \le \nu_\ell \le K$, that proposition gives

$$\gamma_{\rho,K}(\nu_\ell) < A_{\rho,K}(\nu_\ell) + \frac{1}{4}, \tag{3.3}$$

where, after the scaling in (3.1), $A_{\rho,K}(\nu) = \mathcal{A}(\varepsilon\nu)$ and $\mathcal{A}$ is the explicit action in (3.9) below. The exact radial count in channel ℓ is

$$\#\{n \ge 1 : n < \gamma_{\rho,K}(\nu_\ell)\}. \tag{3.4}$$

In particular, (3.3) is an inequality for the strict count; it is not a replacement of that count by an ordinary floor.

3.1 The product comparison

Under the standard unitary radial transformation, the angular channel $\ell \in \mathbb{N}_0$ is the Dirichlet operator

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2} \quad \text{in } L^2((\rho, 1), dr),$$

and it occurs with multiplicity $2\ell + 1$. The following lemma includes the direction of the min–max comparison and every strict radial and angular wall.

Lemma 3.1 (Strict product majorant)**.** *Let $0 < \varepsilon < 1$, $a = \varepsilon K$, and define*

$$N(a) = \max\left\{0, \left\lceil \frac{a}{\pi} \right\rceil - 1\right\}. \tag{3.5}$$

For $1 \le n \le N(a)$, put

$$b_n = \sqrt{a^2 - n^2\pi^2}, \qquad M_n = \left\lceil \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2} \right\rceil.$$

Then

$$N_D(A_{\rho,1}, K^2) \le \mathcal{P}_\varepsilon(a) := \sum_{n=1}^{N(a)} M_n^2. \tag{3.6}$$

In particular, the strict shell count is zero for $0 \le a \le \pi$.

Proof. The two quadratic forms have the common domain $H_0^1(\rho, 1)$, and

$$\int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2}|u|^2\right) dr \ge \int_\rho^1 \left(|u'|^2 + \ell(\ell+1)|u|^2\right) dr,$$

because $r \le 1$. The min–max principle therefore gives

$$\lambda_{\ell,n} \ge \left(\frac{n\pi}{\varepsilon}\right)^2 + \ell(\ell+1), \qquad n \ge 1.$$

Consequently, a shell eigenvalue in this channel can be counted below K^2 only if

$$n^2\pi^2 + \varepsilon^2\ell(\ell+1) < a^2. \tag{3.7}$$

The active radial integers are exactly those satisfying $n\pi < a$, whose number is (3.5). For a fixed active n, condition (3.7) is

$$\ell < \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2}.$$

Hence the allowed angular indices are $0, \dots, M_n - 1$, and their full spherical multiplicity is

$$\sum_{\ell=0}^{M_n-1} (2\ell+1) = M_n^2.$$

Summing proves (3.6).

If $a = m\pi$, the radial level $n = m$ is excluded by the strict inequality $n\pi < a$; thus $N(a) = m - 1$. At an angular wall $b_n^2/\varepsilon^2 = L(L+1)$, the displayed expression inside the ceiling is exactly L, so $M_n = L$ and the level $\ell = L$ is excluded. The jump of multiplicity $2L + 1$ occurs only above the wall. Finally, no positive integer satisfies $n\pi < a$ when $0 \le a \le \pi$, including both endpoints, so the product majorant and hence the shell count are zero there. $\square$

3.2 The complementary-action comparison

For $r \in \{\rho, 1\}$ and $0 \le y \le ar$, define

$$J_r(y) = \sqrt{a^2r^2 - y^2} - y \arccos \frac{y}{ar}, \tag{3.8}$$

and set

$$\mathcal{A}(y) = \begin{cases} \dfrac{J_1(y) - J_\rho(y)}{\pi\varepsilon}, & 0 \le y \le \rho a, \\ \dfrac{J_1(y)}{\pi\varepsilon}, & \rho a \le y \le a. \end{cases} \tag{3.9}$$

We extend $\mathcal{A}(y) = 0$ for $y > a$.

Lemma 3.2 (Action geometry and curvature switch)**.** *The function $\mathcal{A}$ is continuous, continuously differentiable at $y = \rho a$, and strictly decreasing from*

$$T := \mathcal{A}(0) = \frac{a}{\pi} \quad \textit{to} \quad \mathcal{A}(a) = 0.$$

Let $R : [0, T] \to [0, a]$ be its decreasing inverse, let

$$\tau = \mathcal{A}(\rho a), \qquad F(t) = R(t)^2, \qquad g(t) = -F'(t).$$

Then

$$\begin{gathered} g \textit{ decreases on } (0, \tau), \qquad g \textit{ increases on } (\tau, T), \\ F(0) = a^2, \qquad F(\tau) = \rho^2 a^2, \qquad F(T) = 0, \qquad g(T) = 2\pi\rho a, \end{gathered} \tag{3.10}$$

where $g(T)$ denotes the one-sided limit. Moreover $g(t) = O(t^{-1/3})$ as $t \downarrow 0$.

Proof. Direct differentiation on the appropriate open intervals gives

$$J_r'(y) = -\arccos \frac{y}{ar}, \qquad J_r''(y) = \frac{1}{\sqrt{a^2r^2 - y^2}}. \tag{3.11}$$

At $y = \rho a$, $J_\rho(\rho a) = 0$, and the two one-sided first derivatives of (3.9) both equal $-\arccos \rho/(\pi\varepsilon)$. Thus the two pieces and their first derivatives match. Both derivatives are negative away from $y = a$, so the inverse exists on exactly the stated interval.

On the outer branch $\rho a < y < a$, put $\alpha = \arccos(y/a)$. If $t = \mathcal{A}(y)$, then

$$g(t) = -\frac{2y}{\mathcal{A}'(y)} = \frac{2\pi\varepsilon y}{\alpha}. \tag{3.12}$$

The required monotonicity is explicit:

$$\frac{d}{dy}\left(\frac{y}{\alpha}\right) = \frac{\alpha + y/\sqrt{a^2 - y^2}}{\alpha^2} > 0.$$

Since $y = R(t)$ decreases with t, equation (3.12) shows that g decreases on $0 < t < \tau$.

On the inner branch $0 < y < \rho a$, put

$$\delta(y) = \arccos \frac{y}{a} - \arccos \frac{y}{\rho a} > 0.$$

Differentiating $\arccos(y/(ar))$ with respect to r gives the exact identity

$$\frac{\delta(y)}{y} = \int_\rho^1 \frac{dr}{r\sqrt{a^2r^2 - y^2}}. \tag{3.13}$$

The integrand increases strictly with y, so $\delta(y)/y$ increases, and $y/\delta(y)$ decreases, with y. Since

$$g(t) = \frac{2\pi\varepsilon y}{\delta(y)}$$

and $y = R(t)$ decreases, g increases on $\tau < t < T$. Formula (3.12) and its inner counterpart agree at the interface.

The three values of F follow from $R(0) = a$, $R(\tau) = \rho a$, and $R(T) = 0$. As $y \downarrow 0$, (3.13) gives

$$\delta(y) = \frac{\varepsilon}{\rho a} y + O(y^3),$$

and therefore $g(T) = 2\pi\rho a$.

Finally, on the outer branch write $y = a\cos\theta$. Then

$$t = \frac{a}{\pi\varepsilon}(\sin\theta - \theta\cos\theta) \sim \frac{a\theta^3}{3\pi\varepsilon}, \qquad g(t) = \frac{2\pi\varepsilon a\cos\theta}{\theta} = O(t^{-1/3}). \tag{3.14}$$

This proves the asserted improper endpoint behavior. All square roots and arccosines in the proof are taken on their stated closed domains; every division occurs in an open interval where its denominator is positive, and the endpoint zeros are handled by the limits just computed. □

Lemma 3.3 (Strict phase-to-action layer cake)**.** *For $\ell \in \mathbb{N}_0$, set*

$$y_\ell = \varepsilon\left(\ell + \frac{1}{2}\right), \qquad q_\ell = \left[\mathcal{A}(y_\ell) + \frac{1}{4}\right]_<, \qquad P_{\mathcal{A}} = \varepsilon \sum_{\ell \geq 0} y_\ell q_\ell. \tag{3.15}$$

Then

$$N_D(A_{\rho,1}, K^2) \leq \frac{2P_{\mathcal{A}}}{\varepsilon^2}. \tag{3.16}$$

For $n \geq 1$, put $x_n = n - \frac{1}{4}$, and define

$$M_\varepsilon(x) = \#\left\{\ell \in \mathbb{N}_0 : \varepsilon\left(\ell + \frac{1}{2}\right) < x\right\} = \max\left\{0, \left\lceil \frac{x}{\varepsilon} - \frac{1}{2} \right\rceil\right\}. \tag{3.17}$$

The proxy has the exact finite representation

$$P_{\mathcal{A}} = \frac{\varepsilon^2}{2} \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2. \tag{3.18}$$

Moreover,

$$I := \frac{1}{2}\int_0^T F(t)\,dt = \frac{a^3}{3\pi}\left(1 - \varepsilon + \frac{\varepsilon^2}{3}\right), \qquad \frac{2I}{\varepsilon^2} = \frac{2}{9\pi}(1-\rho^3)K^3. \tag{3.19}$$

Proof. For $y_\ell < a$, equations (3.3) and (3.4) show that the number of radial levels in channel ℓ is at most q_ℓ. If $y_\ell \geq a$, equivalently $\nu_\ell \geq K$, the product comparison gives

$$\lambda_{\ell,1} \geq \frac{\pi^2}{\varepsilon^2} + \ell(\ell+1) = \frac{\pi^2}{\varepsilon^2} + \nu_\ell^2 - \frac{1}{4} > K^2,$$

so there is no radial level; at $y_\ell = a$, the definition also gives $q_\ell = [1/4]_< = 0$. Multiplicity $2\ell + 1 = 2y_\ell/\varepsilon$ now yields

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\ell \geq 0} \frac{2y_\ell}{\varepsilon} q_\ell = \frac{2P_{\mathcal{A}}}{\varepsilon^2},$$

which proves (3.16). If $\mathcal{A}(y_\ell) + 1/4 = m \in \mathbb{Z}$, then $q_\ell = m - 1$, so the phase-proxy wall is excluded exactly as required.

For every $y \geq 0$, the strict bracket is

$$\left[\mathcal{A}(y) + \frac{1}{4}\right]_< = \#\{n \geq 1 : x_n < \mathcal{A}(y)\}.$$

The sums are finite. Since $\mathcal{A}$ and R are decreasing, $x_n < \mathcal{A}(y_\ell)$ is equivalent to $y_\ell < R(x_n)$. Therefore

$$\begin{aligned} P_{\mathcal{A}} &= \varepsilon \sum_{\substack{n\geq 1\\ x_n<T}} \sum_{y_\ell < R(x_n)} y_\ell \\ &= \varepsilon \sum_{\substack{n\geq 1\\ x_n<T}} \varepsilon \sum_{\ell=0}^{M_\varepsilon(R(x_n))-1} \left(\ell + \frac{1}{2}\right) \\ &= \frac{\varepsilon^2}{2} \sum_{\substack{n\geq 1\\ x_n<T}} M_\varepsilon(R(x_n))^2, \end{aligned}$$

because $\sum_{\ell=0}^{M-1}(\ell + 1/2) = M^2/2$. This proves (3.18). Formula (3.17) follows from the same strict counting; at $x/\varepsilon = m + 1/2$, it equals m, excluding $\ell = m$.

Differentiation of (3.8) with respect to r gives

$$\frac{\partial J_r(y)}{\partial r} = \sqrt{a^2 - \frac{y^2}{r^2}} \quad (0 \leq y \leq ar).$$

Consequently the two cases of (3.9) combine as

$$\mathcal{A}(y) = \frac{1}{\pi\varepsilon} \int_\rho^1 \left(a^2 - \frac{y^2}{r^2}\right)_+^{1/2} dr, \tag{3.20}$$

where $s_+ = \max\{s, 0\}$. The area identity for a decreasing inverse and Tonelli's theorem give

$$\frac{1}{2}\int_0^T R(t)^2\, dt = \int_0^a y\mathcal{A}(y)\, dy.$$

Using (3.20) and then $y = rz$,

$$\begin{aligned} \int_0^a y\mathcal{A}(y)\, dy &= \frac{1}{\pi\varepsilon}\int_\rho^1 r^2\, dr \int_0^a z\sqrt{a^2 - z^2}\, dz \\ &= \frac{a^3}{3\pi\varepsilon}\int_\rho^1 r^2\, dr = \frac{a^3}{3\pi}\left(1 - \varepsilon + \frac{\varepsilon^2}{3}\right). \end{aligned}$$

This is (3.19); its final equality follows from $1 - \rho^3 = 3\varepsilon m_\varepsilon$ and $K = a/\varepsilon$. □

Lemma 3.4 (Shifted sawtooth and radial quadrature deficit)**.** *Let*

$$C(t) = \#\{n \geq 1 : n - \tfrac{1}{4} < t\}, \qquad w(t) = t - C(t), \qquad \mathfrak{W}(t) = \int_0^t w(s)\, ds,$$

and put $\Psi(t) = \mathfrak{W}(t) - t/4$*. Then, for all* $t \geq 0$*,*

$$\mathfrak{W}(t) \geq 0, \qquad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32}. \tag{3.21}$$

Furthermore, with

$$D_{\mathrm{rad}} = \int_0^T F(t)\,dt - \sum_{\substack{n\ge 1\\ x_n<T}} F(x_n), \tag{3.22}$$

one has

$$D_{\mathrm{rad}} \ge \frac{\rho^2 a^2}{4} - \frac{\pi\rho a}{4}. \tag{3.23}$$

Proof. On $0 \le t \le 3/4$, $C(t) = 0$, and hence

$$\Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \tag{3.24}$$

For $x_n = n - 1/4$, $n \ge 1$, and $t = x_n + u$ with $0 \le u \le 1$, direct integration gives

$$\Psi(t) = \frac{1}{2}\left(u - \frac{1}{2}\right)^2 - \frac{1}{32}. \tag{3.25}$$

These formulas agree at cell endpoints and prove the two-sided bound for Ψ. On the first cell $\mathfrak{W} = t^2/2 \ge 0$. Thereafter $\mathfrak{W} = t/4 + \Psi \ge t/4 - 1/32 > 0$, proving (3.21). At a jump x_n, strict counting gives the one-sided values $w(x_n^-) = 3/4$ and $w(x_n^+) = -1/4$, while $\mathfrak{W}$ and Ψ are continuous.

Distributionally, $dw = dt - dC$. Since $F(T) = 0$, an atom at $x_n = T$ has zero weight and is consistent with its exclusion from (3.22). Stieltjes integration by parts, first on $[\eta, T]$ and then with $\eta \downarrow 0$, gives

$$D_{\mathrm{rad}} = \int_0^T w(t)g(t)\,dt. \tag{3.26}$$

Indeed, (3.14), together with $w(t) = t$ and $\mathfrak{W}(t) = t^2/2$ near zero, gives $w(t)g(t) = O(t^{2/3})$ and $\mathfrak{W}(t)g(t) = O(t^{5/3}) \to 0$, so the improper boundary term vanishes.

Split (3.26) at the actual value τ, with no assumption that τ belongs to any particular sawtooth cell. On $(0, \tau)$, g decreases and $\mathfrak{W} \ge 0$; hence

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_0^\tau \mathfrak{W}\,dg \ge \mathfrak{W}(\tau)g(\tau). \tag{3.27}$$

On (τ, T), g increases and $w = 1/4 + \Psi'$ almost everywhere. Adding its integration-by-parts identity to (3.27), and using $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, cancels the interface value and yields

$$D_{\mathrm{rad}} \ge \frac{F(\tau)}{4} + \frac{\tau g(\tau)}{4} + \Psi(T)g(T) - \int_\tau^T \Psi\,dg.$$

Because dg is a positive measure on (τ, T), the bounds (3.21) imply

$$\begin{aligned} D_{\mathrm{rad}} &\ge \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) \\ &\ge \frac{F(\tau)}{4} - \frac{g(T)}{8}. \end{aligned}$$

The two discarded quantities, $-\int_0^\tau \mathfrak{W}\,dg$ and $g(\tau)(\tau/4 + 3/32)$, are explicitly nonnegative. Substitution of $F(\tau) = \rho^2 a^2$ and $g(T) = 2\pi\rho a$ from Lemma 3.2 proves (3.23). Continuity of $\mathfrak{W}, \Psi, g$ at the split shows that the same proof covers τ below, on, or above every sawtooth grid point. $\square$

4 Global two-ratio closure

4.1 Ratio-sharp defect theorem below the optical strip

For $a > 0$, let

$$G_a(z) := \frac{1}{\pi}\left(\sqrt{a^2 - z^2} - z\arccos\frac{z}{a}\right) \quad (0 \le z < a), \qquad G_a(z) := 0 \quad (z \ge a),$$

and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := \frac{(1-\rho)K}{\pi}.$$

The strict phase reduction and the inverse-action layer cake give

$$N_D(A_{\rho,1}, K^2) \le P, \qquad P = \sum_{n-1/4<T} M(R(n-1/4))^2, \qquad W = \int_0^T R(t)^2\,dt, \tag{4.1}$$

where R is the decreasing inverse of A and $M(r)$ counts half-integers strictly below r. Thus

$$W - P = \mathcal{D}_{\mathrm{rad}} - \mathcal{E}_{\mathrm{ang}}.$$

Write $\theta = \arccos\rho$ and $N = \#\{n \ge 1 : n - 1/4 < T\}$. The exact action derivative satisfies $-A' \le \theta/\pi$. One disjoint left block per active layer therefore gives the ratio-sharp estimate

$$\mathcal{E}_{\mathrm{ang}} < \frac{(1-\rho^2)K^2}{6} - N\left(\frac{3\pi}{8\theta} - \frac{1}{4}\right). \tag{4.2}$$

Strictness includes a half-integer angular wall.

For $0 < \rho \le 39/50$ and $K \ge \pi/(1-\rho)$, the curvature interface $\tau = KG_1(\rho)$ satisfies $\tau > 1/4$. Replacing the shifted-sawtooth primitive by its C^1 tangent envelope

$$L_\sharp(t) = \begin{cases} t^2/2, & t \le 1/4, \\ t/4 - 1/32, & t \ge 1/4, \end{cases}$$

and retaining the decreasing-branch Stieltjes remainder gives

$$\mathcal{D}_{\mathrm{rad}} \ge B_0(K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{\pi\rho K}{4\arccos\rho}, \qquad B_0(K) := \int_0^{1/4} R_B(t)^2\,dt, \tag{4.3}$$

where R_B is the inverse ball action. The turning estimate $G_1(s) \ge 2\sqrt{2}(1-s)^{3/2}/(3\pi)$ yields

$$B_0(K) \ge \frac{K^2}{4} - \alpha K^{4/3} + \beta K^{2/3}, \quad \alpha = \frac{6C}{5\,4^{5/3}}, \quad \beta = \frac{3C^2}{7\,4^{7/3}}, \quad C = \frac{(3\pi)^{2/3}}{2}. \tag{4.4}$$

Since $N \ge 1$ and $\arccos\rho \le (\pi/2)\sqrt{1-\rho}$, it remains to prove positivity of

$$\begin{aligned}\underline{\mathcal{G}}(\rho, K) := \frac{1+2\rho^2}{12}K^2 - \alpha K^{4/3} + \beta K^{2/3} - \frac{\pi\rho K}{4(1-\rho)} \\ + \frac{\rho K}{2\sqrt{1-\rho}} + \frac{3}{4\sqrt{1-\rho}} - \frac{1}{4}.\end{aligned} \tag{4.5}$$

On the base $K = \pi/(1-\rho)$, the substitution $x = (1-\rho)^{-1/6}$ reduces (4.5) to one explicit degree-nine expression. The detailed support module proves it positive from $157/50 < \pi < 22/7$, positive-side cubing, and

$$F(1) > \frac{1}{4}, \qquad F'(1) > -\frac{5}{2}, \qquad F''(x) > 14.$$

It also proves that the base K-derivative is positive and

$$\partial_K^2 \underline{\mathcal{G}} > \frac{47447}{443682} > 0.$$

These are finite exact inequalities, not sampled or interval premises.

Theorem 4.1 (Global low- and middle-ratio theorem)**.** *For*

$$0 < \rho \leq \frac{39}{50}, \qquad K \geq \frac{\pi}{1-\rho},$$

one has

$$N_D(A_{\rho,1}, K^2) \leq P < W(\rho, K).$$

Proof. Equations (4.2)–(4.4) give

$$W - P > \underline{\mathcal{G}}(\rho, K).$$

The exact scalar closure just described proves the right side positive on the stated domain. All intermediate identities and the rational convexity proof are given in the accompanying analytic supplement. □

Lemma 4.2 (Rational bounds for π)**.** *One has*

$$\frac{157}{50} < \frac{333}{106} < \pi < \frac{22}{7}, \qquad \textit{and hence} \qquad \frac{1}{\pi} < \frac{106}{333}.$$

Proof. Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. The tangent addition formula gives

$$\tan(2a) = \frac{5}{12}, \qquad \tan(4a) = \frac{120}{119}, \qquad \tan(4a - b) = 1.$$

Here $b < a$, while $(6/5)^2 < 2$ gives $1/5 < \sqrt{2} - 1 = \tan(\pi/8)$, and hence $a < \pi/8$. Therefore $0 < 4a - b < \pi/2$. Thus this is Machin's identity $\pi = 16a - 4b$. The alternating arctangent expansion gives

$$a > \frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5} - \frac{1}{7\,5^7} = \frac{323852}{1640625}, \qquad b < \frac{1}{239}.$$

Indeed, these inequalities follow by integrating

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \frac{t^8}{1+t^2} \qquad (0 \leq t \leq 1)$$

and using $1/(1+t^2) < 1$ for $t > 0$. Therefore

$$\pi > \frac{1231847548}{392109375} = \frac{333}{106} + \frac{3418213}{41563593750}.$$

Also $333/106 - 157/50 = 2/1325 > 0$. For the upper bound,

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2},$$

so integration and $4 \arctan 1 = \pi$ give

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\, dx = \frac{22}{7} - \pi.$$

Since $22/7 - 333/106 = 1/742 > 0$, all four rational numbers are ordered as asserted. This proves every displayed comparison. □

4.2 All-frequency optical theorem

Lemma 4.3 (All-frequency optical theorem)**.** *For* $39/50 \le \rho < 1$ *and* $K \ge 0$*,* $N_D(A_{\rho,1}, K^2) \le W(\rho, K)$*, with strict inequality for* $K > 0$*.*

Proof. Put $\varepsilon = 1 - \rho \le 11/50$, $a = \varepsilon K$, and $c = 1126/625$. We use two inclusive screens meeting at $a = c/\varepsilon$.

For the product screen, the strict min–max condition (3.7) implies that a counted mode satisfies

$$n^2\pi^2 + \varepsilon^2 \ell(\ell+1) < a^2.$$

There is no mode for $a \le \pi$. For $a > \pi$, write $a/\pi = N + t$, with $N \ge 1$, $0 < t \le 1$, assigning $(N, t) = (m-1, 1)$ at $a = m\pi$. If

$$b_n = \sqrt{a^2 - n^2\pi^2}, \qquad M_n = \left\lceil \sqrt{(b_n/\varepsilon)^2 + 1/4} - 1/2 \right\rceil,$$

the exact angular multiplicity is bounded by

$$\mathcal{P}_\varepsilon(a) = \sum_{n=1}^{N} M_n^2.$$

Set

$$I_0(a) = \int_0^{a/\pi} (a^2 - \pi^2 x^2)\, dx = \frac{2a^3}{3\pi}, \qquad S_0(a) = \sum_{n=1}^{N} b_n^2, \qquad D(a) = I_0(a) - S_0(a).$$

Direct summation of the continuous-minus-discrete product deficit gives

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{4.6}$$

For $C_D = 1382/3125$, the difference $H(N, t) := D/\pi^2 - C_D(N + t)^2$ increases in N. Indeed, for $N \ge 1$ and $0 < t \le 1$,

$$\partial_N H = (1 - 2C_D)N + t^2 + \frac{1}{6} - 2C_D t \ge 1 - 2C_D + \frac{1}{6} - C_D^2 = \frac{5074831}{58593750} > 0.$$

At $N = 1$, one has

$$\partial_t H(1, t) = 2(t - C_D)(1 + t),$$

so the minimum occurs at $t = C_D$, where it equals $1822532/91552734375 > 0$. Thus $D > C_D a^2$.

For $x > 0$,

$$\left\lceil \sqrt{x^2 + \frac{1}{4}} - \frac{1}{2} \right\rceil^2 < x^2 + x + 1.$$

Indeed the ceiling is strictly smaller than $\sqrt{x^2 + 1/4} + 1/2$, whose square is $x^2 + 1/2 + \sqrt{x^2 + 1/4} < x^2 + x + 1$. Applying this with $x = b_n/\varepsilon$, then using the monotone left-sum estimate and the quarter-circle integral,

$$\sum_{n=1}^{N} b_n < \int_0^{a/\pi} \sqrt{a^2 - \pi^2 x^2}\, dx = \frac{a^2}{4}, \qquad N < \frac{a}{\pi},$$

gives

$$\varepsilon^2 \mathcal{P}_\varepsilon < S_0 + \varepsilon a^2/4 + \varepsilon^2 a/\pi.$$

On $a \le c/\varepsilon$, using $q = 106/333 > 1/\pi$ from Lemma 4.2 and $\varepsilon \le 11/50$, the exact reserve is

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + (11/50)^2 q^2\right) = \frac{39569}{2772225000} > 0. \tag{4.7}$$

Indeed, $a \le c/\varepsilon$, $\varepsilon \le 11/50$, $a > \pi$, and $q > 1/\pi$ give

$$\varepsilon I_0 + \frac{\varepsilon a^2}{4} + \frac{\varepsilon^2 a}{\pi} < a^2\left(\frac{2cq}{3} + \frac{11}{200} + (11/50)^2 q^2\right) < C_D a^2 < D.$$

Consequently

$$\varepsilon^2 \mathcal{P}_\varepsilon < I_0 - D + \frac{\varepsilon a^2}{4} + \frac{\varepsilon^2 a}{\pi} < (1-\varepsilon) I_0 < m_\varepsilon I_0 = \varepsilon^2 W,$$

where the last identity is (3.2). This proves the low screen, including all radial and angular equality walls.

For the complementary-action screen, use the inverse-action construction proved in Lemma 3.2: let the decreasing shell action have inverse $R_{\mathcal{A}}$, put $F = R_{\mathcal{A}}^2$, and $g = -F'$. If τ is the actual curvature interface and $T = a/\pi$, that lemma gives

$$g \text{ decreasing on } (0,\tau), \quad g \text{ increasing on } (\tau, T),$$

$$F(0) = a^2, \quad F(\tau) = \rho^2 a^2, \quad F(T) = 0, \quad g(T) = 2\pi\rho a.$$

For $C(t) = \#\{n : n - 1/4 < t\}$, set $w = t - C(t)$ and $\Psi(t) = \int_0^t w(s)\,ds - t/4$. The complete sawtooth and Stieltjes calculation in Lemma 3.4, including the ungridded curvature interface τ, gives

$$D_{\mathrm{rad}} := \int_0^T F(t)\,dt - \sum_{n-1/4<T} F(n-1/4) \ge \frac{\rho^2 a^2}{4} - \frac{\pi\rho a}{4}. \tag{4.8}$$

The exact half-integer angular count from Lemma 3.3 satisfies $\varepsilon^2 M_\varepsilon(x)^2 < (x+\varepsilon/2)^2$, even on an angular wall. Since fewer than $a/\pi + 1/4$ radial layers occur, the angular error is strictly below

$$E = \left(\frac{a}{\pi} + \frac{1}{4}\right)\left(\varepsilon a + \frac{\varepsilon^2}{4}\right). \tag{4.9}$$

On $a \ge c/\varepsilon$, one has $1/a \le \varepsilon/c$. Put

$$L_{\mathrm{opt}}(\varepsilon) := \frac{(1-\varepsilon)^2}{4} - \frac{(22/7)(1-\varepsilon)\varepsilon}{4c},$$

$$A_{\mathrm{opt}}(\varepsilon) := q\varepsilon + \frac{\varepsilon^2}{4c} + \frac{q\varepsilon^3}{4c} + \frac{\varepsilon^4}{16c^2}.$$

Dividing (4.8) and (4.9) by a^2, then using Lemma 4.2, gives

$$\frac{D_{\mathrm{rad}}}{a^2} \ge L_{\mathrm{opt}}(\varepsilon), \qquad \frac{E}{a^2} < A_{\mathrm{opt}}(\varepsilon).$$

For $0 < \varepsilon \le 11/50 < 1/2$,

$$L'_{\mathrm{opt}}(\varepsilon) = -\frac{1-\varepsilon}{2} - \frac{22/7}{4c}(1-2\varepsilon) < 0, \qquad A'_{\mathrm{opt}}(\varepsilon) > 0.$$

At $\varepsilon = 11/50$, the exact endpoint difference is

$$L_{\mathrm{opt}}(11/50) - A_{\mathrm{opt}}(11/50) = \frac{14817541}{472867032960000} > 0. \tag{4.10}$$

Thus the strict layer-cake sum is below the action integral, and the phase bridge proves the high screen. The common face lies in the nonzero radial range because

$$\frac{c}{\varepsilon} \geq \frac{c}{11/50} = \frac{2252}{275} > \frac{22}{7} > \pi, \qquad \frac{2252}{275} - \frac{22}{7} = \frac{9714}{1925} > 0.$$

Both screens include $a = c/\varepsilon$, so their union has neither a gap nor an unowned equality face. At $K = 0$ both sides vanish. $\square$

Exact optical reserve ledger. The small margins in the preceding proof are recorded here to make later constant changes auditable. Each entry is an exact positive rational quantity, not a decimal or numerical premise.

certificate	positive reserve
Machin lower certificate above 333/106	3418213/41563593750
minimum of the product-deficit function H	1822532/91552734375
low optical screen	39569/2772225000
high optical screen	14817541/472867032960000
meeting face above 22/7	9714/1925

5 Proof of the spectral theorem

We now assemble the three disjoint phase-space owners. Fix $0 < \rho < 1$ and put

$$K_0(\rho) := \frac{\pi}{1-\rho}.$$

If $0 \leq K \leq K_0(\rho)$, the strict product majorant in Lemma 3.1 gives

$$N_D(A_{\rho,1}, K^2) = 0. \tag{5.1}$$

This includes $K = K_0(\rho)$: the first one-dimensional Dirichlet frequency lies on the spectral wall and is excluded.

It remains to take $K > K_0(\rho)$. If $0 < \rho < 39/50$, apply Theorem 4.1. If $39/50 \leq \rho < 1$, apply Lemma 4.3. These cases, together with (5.1), are disjoint and exhaustive. Therefore

$$N_D(A_{\rho,1}, K^2) \leq \frac{2}{9\pi}(1-\rho^3)K^3 \qquad (K \geq 0). \tag{5.2}$$

Now

$$\omega_3 = \frac{4\pi}{3}, \qquad L_3 = \frac{\omega_3}{(2\pi)^3} = \frac{1}{6\pi^2}, \qquad |A_{\rho,1}| = \frac{4\pi}{3}(1-\rho^3),$$

so

$$L_3|A_{\rho,1}| = \frac{2}{9\pi}(1-\rho^3). \tag{5.3}$$

Taking $K = \sqrt{\Lambda}$ in (5.2) proves the unit-shell form of Theorem 1.1 for every $\Lambda \geq 0$.

For arbitrary $0 < r < R$, put $\rho = r/R$. The unitary dilation

$$(D_R u)(x) = R^{-3/2} u(x/R)$$

maps $L^2(A_{\rho,1})$ onto $L^2(A_{r,R})$, and its Dirichlet form scales by R^{-2}. Hence

$$\lambda_j^D(A_{r,R}) = R^{-2}\lambda_j^D(A_{\rho,1}),$$

and the strict threshold transforms as

$$N_D(A_{r,R}, \Lambda) = N_D(A_{\rho,1}, R^2\Lambda). \tag{5.4}$$

Applying the unit-shell result at $R^2\Lambda$,

$$\begin{aligned} N_D(A_{r,R}, \Lambda) &\le \frac{2}{9\pi}(1-\rho^3)(R^2\Lambda)^{3/2} \\ &= \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} \\ &= L_3|A_{r,R}|\Lambda^{3/2}. \end{aligned}$$

This proves Theorem 1.1 for every $R > 0$, including $R < 1$.

A Independent non-tiling theorem

We prove Theorem 1.2 independently of the spectral argument. Set $\alpha = r/R \in (0, 1)$ and scale by $1/R$. It suffices to consider

$$T = A_{\alpha,1} = \{x : \alpha < |x| < 1\}.$$

Every rigid motion has the form $x \mapsto Qx + a$, with Q orthogonal. Because T is radial, $Q(T) = T$, so every congruent copy is a translate

$$T_a = a + T = \{x : \alpha < |x - a| < 1\}.$$

Assume for contradiction that $\{T_a : a \in \mathfrak{C}\}$ has pairwise-disjoint interiors and covers almost everywhere. Duplicate centers are impossible because they give the same nonempty open tile.

Fix a unit vector e and put

$$c = \frac{1+\alpha}{2}e, \qquad \delta = \frac{1-\alpha}{2}.$$

Write $B(x, s) = \{y \in \mathbb{R}^3 : |y - x| < s\}$ for the open Euclidean ball. Then $B(c, \delta) \subset T$. Therefore T_a contains $B(a + c, \delta)$, and disjointness implies

$$|a - b| \ge 1 - \alpha \qquad (a \ne b). \tag{A.1}$$

Thus the centers are locally finite: a bounded set cannot contain infinitely many points with fixed positive separation. They are also countable, since

$$\mathfrak{C} = \bigcup_{m \ge 1} (\mathfrak{C} \cap \overline{B}(0, m))$$

is a countable union of finite sets.

Each tile boundary is a union of two spheres and is three-dimensional null. The countable union

$$Z = \bigcup_{a \in \mathfrak{C}} \partial T_a$$

is therefore null. Exact open-shell coverage is already a special case of almost-everywhere coverage. If closed shells cover exactly or almost everywhere, removing Z shows that their open interiors still cover almost everywhere. Hence it is enough to assume

$$\left| \mathbb{R}^3 \setminus \bigcup_{a \in \mathfrak{C}} T_a \right| = 0. \tag{A.2}$$

Choose one tile T_{a_0} and its outer sphere

$$\Sigma = \{x : |x - a_0| = 1\}.$$

Fix $p \in \Sigma$ and put $u = p - a_0$. For $n \geq 1$, let

$$U_n = B\left(p + \frac{2}{n}u, \frac{1}{n}\right).$$

Every U_n lies outside T_{a_0}, is open, and has positive volume. By (A.2), choose $x_n \in U_n \cap T_{a_n}$ with $a_n \neq a_0$. Since $x_n \to p$ and $|x_n - a_n| < 1$, the centers a_n eventually lie in a fixed bounded set. Local finiteness lets us pass to a subsequence with one fixed center $a \neq a_0$. Thus $p \in \overline{T_a}$.

The point p is not in T_a. Otherwise some $B(p, \varepsilon)$ lies in T_a. Choose

$$0 < t < \min\{\varepsilon, 1 - \alpha\}.$$

Then $p - tu \in T_{a_0} \cap T_a$, contradicting disjoint interiors. Therefore every $p \in \Sigma$ belongs to the boundary of a neighboring tile.

If $p \in \Sigma \cap \partial T_a$, then $|p - a| \in \{\alpha, 1\}$, hence $|a - a_0| \leq 2$. Only finitely many centers satisfy this bound. Consequently Σ is covered by finitely many neighboring inner or outer boundary spheres.

Translate so that $a_0 = 0$. The intersection of Σ with a neighboring sphere $\{x : |x-a| = s\}$, $s \in \{\alpha, 1\}$, satisfies

$$|x|^2 = 1, \qquad |x - a|^2 = s^2,$$

and hence

$$2a \cdot x = |a|^2 + 1 - s^2. \tag{A.3}$$

Because $a \neq 0$, (A.3) is a genuine plane. Its intersection with Σ is empty, a tangent point, or a circle, and therefore has zero surface measure on Σ. A finite union of such sets cannot cover Σ, whose area is 4π.

There is no coincident-sphere exception. A neighboring outer sphere equals Σ only when it has the same center, which is the original tile or a forbidden duplicate. A neighboring inner sphere cannot equal Σ because $\alpha < 1$. We have reached a contradiction. Scaling back proves Theorem 1.2.

B External inputs and verification boundary

The proof above is self-contained with respect to every shell-specific spectral reduction, weighted count, regional subtraction, boundary assignment, and exact-rational inequality. It uses only the following conventional external results, each in the scope stated where it enters the proof:

1. standard spherical-harmonic decomposition, regular Dirichlet Sturm–Liouville theory, direct-sum facts, and the min–max principle;
2. the Bessel equation and standard identities, its Wronskian, Nicholson's representation, and the usual small- and large-argument formulas collected in [3];
3. the real-order phase enclosure of [1] and the annular phase-difference estimates stated as Theorem 5.1 and Lemma 5.2 of [2].

The rational bounds $157/50 < 333/106 < \pi < 22/7$, including the reciprocal inequality used in the optical theorem, are proved in Lemma 4.2 rather than imported. Every nonstandard scalar estimate is likewise proved in the paper or the accompanying *Purely Analytic Supplement*. No numerical experiment, floating-point computation, interval certificate, or executable program is a premise of the proof.

References

[1] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Uniform enclosures for the phase and zeros of Bessel functions and their derivatives, *SIAM J. Math. Anal.* **56** (2024), no. 6, 7644–7682. doi:10.1137/24M1642032.

[2] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Pólya's conjecture for Dirichlet eigenvalues of annuli, *J. Lond. Math. Soc.* **113** (2026), no. 2, e70425. doi:10.1112/jlms.70425.

[3] NIST Digital Library of Mathematical Functions, Chapter 10: Bessel functions, in particular equation (10.9.30) and the standard identities and limiting formulas used above, `https://dlmf.nist.gov/10`, accessed 16 July 2026.